



INSTALLATION PROJECT 2

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ARTI 405



● Project Overview

Goal: - Deploy a usable, custom machine learning model with a rich user interface

Method: - I am going to be using Streamlit for my interface and deployment. Streamlit is sort of an all-in-one framework for small applications featuring machine learning models. There are functions built into the wrapper that will translate Python code directly into HTML.

Model: - I am going to be using a sentiment analysis model that I created, which is similar to one that was created in ARTI 330.

● Deployment Method

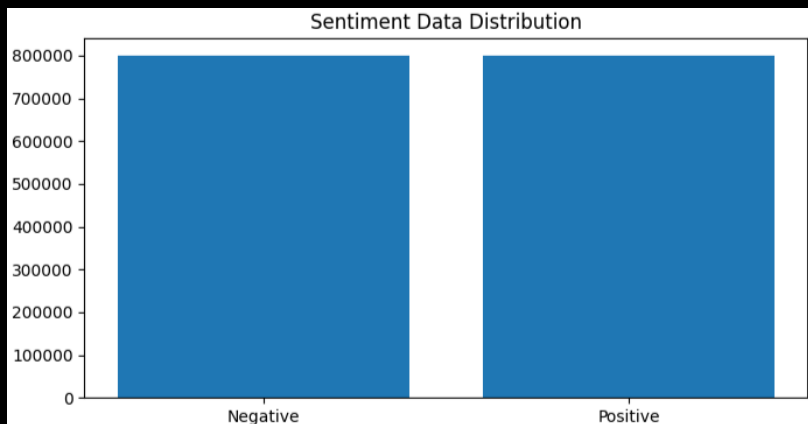
As I said in the previous slide, I will be using Streamlit to manage the interface and deployment. I decided to use Streamlit, rather than Docker, due to its similarity to frameworks that I am already familiar with. I have a lot of experience with Flask and Django in Python, so I'm simply more comfortable with deployment being managed entirely in my Python environment, rather than having a separate thing to manage it, like Docker.

I am also able to easily interchange models whenever I want, and even implement more than one into the application. This is extremely easy to do, because everything is managed within the Python environment, so the only thing I need to do in order to change this, is by changing a few lines of the code.

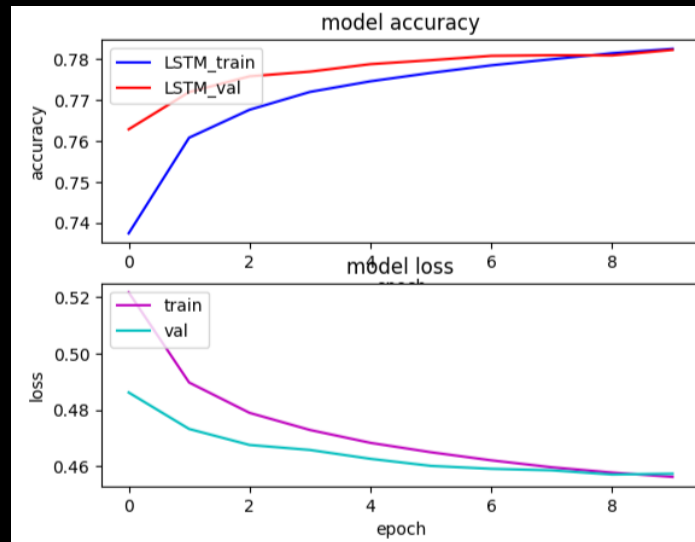


Model Overview

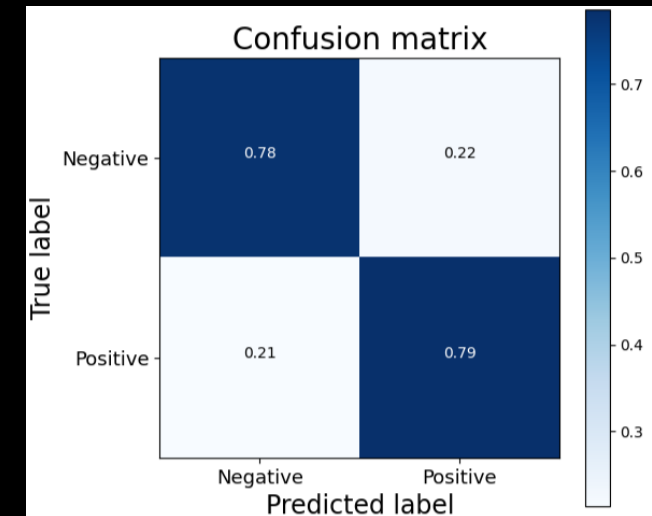
The model I'm using was created to do sentiment analysis on Tweets. For training I used a Kaggle Tweet dataset with ~1.6 million samples, which should be more than enough for decent results.



The dataset was evenly split between negative and positive samples, so the training should be reasonably robust, at least for the purpose of this project.



Overall, the model's performance might seem somewhat low, but for the purpose of sentiment analysis, using only text, this is quite impressive.



The model's missed predictions aren't heavily skewed towards positives or negatives, which is also a huge positive, given the ambiguity of some samples.

● Installation (Pt. 1)

Requirements:

- *Tensorflow*
- *Streamlit*
- *Joblib*
- *Pickle*
- *Pandas*
- *Matplotlib*
- *Wordcloud*

Folder Structure:

Project_Folder
---- *app.py*
---- *Models []*
 ---- *Model*
 ---- *Tokenizer*

There is not a lot of preparation needed, aside from installation of the required packages, and the acquiring of a sentiment model and tokenizer, which are included in this project.

● Installation (Pt. 2)

Due to the plug-and-play nature of Streamlit, so long as the code within the project submission is used, and the required packages are installed, there should be no issues at play.

Run project

```
> streamlit run app.py
```

*Localhost immediately
available*

You can now view your Streamlit app in your browser.

Local URL: <http://localhost:8501>

Network URL: <http://192.168.1.116:8501>

Interface



The screenshot shows a web application titled "Tweet Sentiment Analyzer". It features a text input field for entering a tweet, followed by an "Analyze" button. Below this, there is a section for uploading a CSV file, with a "Browse files" button. The interface is clean and modern, with a dark theme.

Tweet Sentiment Analyzer

Enter a tweet for sentiment analysis:

Analyze

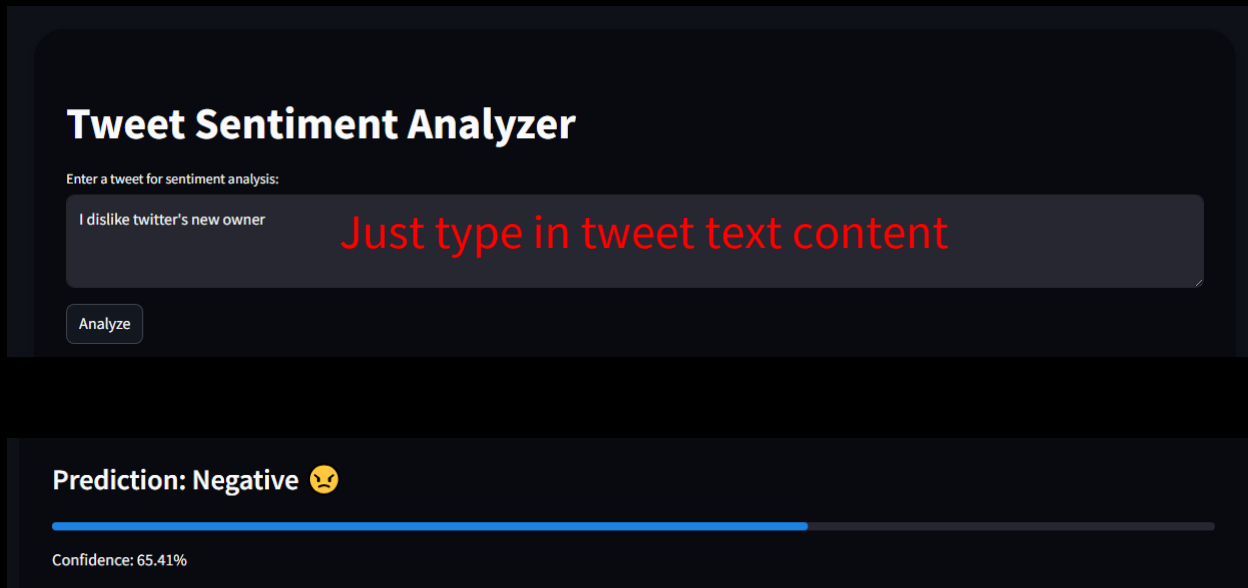
Or upload a CSV file of tweets to analyze:

Upload CSV

Drag and drop file here
Limit 200MB per file • CSV

Browse files

● Example Usage (Pt. 1)



Tweet Sentiment Analyzer

Enter a tweet for sentiment analysis:

I dislike twitter's new owner

Just type in tweet text content

Analyze

Prediction: Negative 😞

Confidence: 65.41%

**Gives “positive” or “negative”
determination, with confidence weight**

There are two options for usage with the application. The first is with single tweets, or test tweet text content. These can simply be typed into the text form in the middle of the page, followed by the pressing of the “Analyze” button.

The resulting prediction is then shown directly below, with a confidence bar and percentage.

● Example Usage (Pt. 2)

Or upload a CSV file of tweets to analyze:

Upload CSV

Drag and drop file here
Limit 200MB per file • CSV

Browse files

Results are displayed in a table below

Prediction Results:

	text	Prediction	Confidence Bar
0	@switchfoot http://twitpic.com/2y1zl - Awww, that's a bummer. You shoulda got David	Negative 😞	(76.8%)
1	is upset that he can't update his Facebook by texting it... and might cry as a result Scho	Negative 😞	(98.7%)
2	@Kenichan I dived many times for the ball. Managed to save 50% The rest go out of box	Negative 😞	(62.0%)
3	my whole body feels itchy and like its on fire	Negative 😞	(94.1%)
4	@nationwideclass no, it's not behaving at all. I'm mad. why am i here? because I can't s	Negative 😞	(73.7%)

You can download your predictions as a CSV below:

Download Predictions as CSV

Results can also be
separately
downloaded

The other option is bulk analysis, which is done by submitting a csv file of tweets, wherein one of the columns should be named “text”, which should contain only the tweet’s text content.

The results will then be displayed below the submission box, in the form of a table, which shows each individual sample’s prediction, along with a confidence percentage and a helpful pseudo-confidence bar